

Bioavailability study of a polyphenol-enriched extract from Hibiscus sabdariffa in rats and associated antioxidant status.

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The aqueous extracts of Hibiscus sabdariffa have been commonly used in folk medicine. Nevertheless, the compounds or metabolites responsible for its healthy effects have not yet been identified. The major metabolites present in rat plasma after acute ingestion of a polyphenol-enriched Hibiscus sabdariffa extract were characterized and quantified in order to study their bioavailability. The antioxidant status of the plasma samples was also measured through several complementary antioxidant techniques. High-performance liquid chromatography coupled to time-of-flight mass spectrometry (HPLC-ESI-TOF-MS) was used for the bioavailability study. The antioxidant status was measured by ferric reducing ability of plasma method, thiobarbituric acid reactive substances assay, and superoxide dismutase activity assay. Seventeen polyphenols and metabolites have been detected and quantified. Eleven of these compounds were metabolites. Although phenolic acids were found in plasma without any modification in their structures, most flavonols were found as quercetin or kaempferol glucuronide conjugates. Flavonol glucuronide conjugates, which show longer half-life elimination values, are proposed to contribute to the observed lipid peroxidation inhibitory activity in the cellular membranes. By contrast, phenolic acids appear to exert their antioxidant activity through ferric ion reduction and superoxide scavenging at shorter times. We propose that flavonol-conjugated forms (quercetin and kaempferol) may be the compounds responsible for the observed antioxidant effects and contribute to the healthy effects of H. sabdariffa polyphenolic extract. © 2012 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.